

Electromagnetic waves* Classification of media:

Single

Continuous media

lossless media(zero conductivity, maintains const. Amp)
No loss of power(Non zero σ signal decay dissipate E into heat)
loss media:

Perfect dielectric

$$\sigma = 0, \epsilon = \epsilon_r \epsilon_0$$

$$\mu = \mu_0$$

Free Space

$$\sigma = 0, \epsilon = \epsilon_0$$

$$\mu = \mu_0$$

Good dielectric

$$\sigma \neq 0, \epsilon = \epsilon_r \epsilon_0$$

$$\mu = \mu_0 \mu_r$$

Good Conductor

$$\sigma = \infty, \epsilon = \epsilon_0$$

$$\mu = \mu_0 \mu_r$$

Perfect conductor

$$\sigma = \infty, \epsilon = \epsilon_0, \mu = \mu_0 \mu_r$$

* Wave equations:

$$\nabla \times \vec{E} = -\dot{\vec{B}}$$

$$\nabla \times \vec{H} = \dot{\vec{D}}$$

$$\nabla \cdot \vec{D} = 0, \quad \nabla \cdot \vec{B} = 0$$

→ we know,

$$\nabla \times \vec{H} = \dot{\vec{D}}$$

• diff. partially w.r.t. t on both sides:

$$\therefore \nabla \times \frac{\partial}{\partial t} \vec{H} = \frac{\partial}{\partial t} \dot{\vec{D}}$$

$$\therefore \nabla \times \dot{\vec{H}} = \ddot{\vec{D}} \longrightarrow$$

$$\text{Similarly, } \nabla \times \dot{\vec{E}} = -\ddot{\vec{B}} \longrightarrow$$

These are wave eqns. for electric & magnetic field.

(A) wave eqn. for $\vec{E}$:

- we know,

$$\nabla \times \vec{E} = -\dot{\vec{B}}$$

curl of both sides :

$$\therefore \nabla \times \nabla \times \vec{E} = -\nabla \times \dot{\vec{B}}$$

$$= -\nabla \times \mu \dot{\vec{H}}$$

$$= -\mu (\nabla \times \dot{\vec{H}})$$

$$= -\mu (\ddot{\vec{B}})$$

$$= -\mu (\epsilon \ddot{\vec{E}}) = -\mu \epsilon \ddot{\vec{E}}$$

$$\therefore \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu \epsilon \ddot{\vec{E}}$$

- but, $\nabla \cdot \vec{E} = 0$

$$\therefore \nabla^2 \vec{E} = \mu \epsilon \ddot{\vec{E}}$$

(B) wave eqn. for $\vec{H}$:

- we know,

$$\nabla \times \vec{H} = \dot{\vec{B}}$$

curl of both sides :

$$\nabla \times \nabla \times \vec{H} = \nabla \times \dot{\vec{B}}$$

$$= \nabla \times \epsilon \dot{\vec{E}}$$

$$= \epsilon (\nabla \times \dot{\vec{E}})$$

$$= \epsilon (-\dot{\vec{B}}) = -\epsilon (\mu \dot{\vec{H}})$$

$$\therefore \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H} = -\mu \epsilon \ddot{\vec{H}}$$

- but, $\nabla \cdot \vec{H} = 0$

$$\therefore \nabla^2 \vec{H} = \mu \epsilon \ddot{\vec{H}}$$

✱

①

locus of points having same phase

- propagation const. $(\gamma) = \text{attenuation const. } (\alpha) + j(\text{phase const.}) (\beta)$

Phase angle = $\beta \times \text{distance}$

here : $\beta = \frac{2\pi}{\lambda}$

(§) Intrinsic Impedance :

Ratio of magnitude of $\vec{E}$ to that of $\vec{M}$ is

$$\text{ie. } \eta = \frac{|E|}{|H|} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}}$$

$\therefore$ in free space

$$\therefore \mu = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi(10^{-7})}{8.85(10^{-12})}} = 377 \, \Omega \text{ or } (120\pi) \, \Omega$$

* Helmholtz Eqn. using Maxwell. eqn. :

a) HE for $\bar{E}$:

we know, $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -j\omega \vec{B}$ — (I)

also, $\nabla \times \vec{H} = j\omega \vec{D} = j\omega (\epsilon \vec{E})$ — (II)

also, $\nabla \cdot \vec{D} = 0 \Rightarrow \nabla \cdot \vec{E} = 0$ — (III)

$$\nabla \cdot \vec{B} = 0 \Rightarrow \nabla \cdot \vec{H} = 0 \quad \text{--- (IV)}$$

Taking curl of eqn. (I) :

$$\begin{aligned}\nabla \times \nabla \times \vec{E} &= \nabla \times (-j\omega \mu \vec{H}) \\ &= -j\omega \mu (\nabla \times \vec{H}) \\ &= -j\omega \mu (j\omega \epsilon \vec{E}) \quad \text{--- (from (II))} \\ \therefore \nabla \times \nabla \times \vec{E} &= \omega^2 \mu \epsilon \vec{E} \quad \text{--- (V)}\end{aligned}$$

Now,

$$\begin{aligned}\nabla \times \nabla \times \vec{E} &= \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} \\ &= -\nabla^2 \vec{E} \quad \text{--- (from (III))} \quad \text{(VI)}\end{aligned}$$

$$\begin{aligned}\therefore -\nabla^2 \vec{E} &= \omega^2 \mu \epsilon \vec{E} \quad \text{--- (from (V) and (VI))} \\ \therefore \nabla^2 \vec{E} &= -\omega^2 \mu \epsilon \vec{E}\end{aligned}$$

Q]

→ Velocity of wave

consider wave is travelling in x -dim. and cosine in nature

$\therefore \cos(\omega t - \beta x)$: constant

$\therefore \omega t - \beta x$: const.

diff. w.r.t. 't' :

$\therefore \omega - \beta \frac{dx}{dt} = 0$

$\therefore \frac{dx}{dt} = \frac{\omega}{\beta} \rightarrow$ velocity of wave

also, $\beta = \omega \sqrt{\mu \epsilon}$

$\therefore \frac{dx}{dt} = \frac{1}{\sqrt{\mu \epsilon}}$

→ $\lambda = \frac{2\pi}{\beta} \rightarrow$ wavelength.

dist. over which sine waves passes ^{full cycle of} 2π radian.

also, $\lambda = \frac{v}{f} = \frac{\text{velo.}}{\text{freq.}}$

→ for lossless media:

$\beta = \omega \sqrt{\mu \epsilon}$

$\alpha = 0$

$\lambda = \frac{2\pi}{\beta} = \frac{v}{f}$

$v = \omega / \beta$

$= \frac{1}{\sqrt{\mu \epsilon}}$

$\eta = \sqrt{\mu / \epsilon}$

any circle $\rightarrow$ $\hat{z} \leftarrow \hat{y}$ anti clockwise: $-\hat{y}$

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q) In free space, $E(x,t) = 10^3 \sin(\omega t - \beta x) \hat{y}$ (V/m)
obtain $H(x,t)$.

$\rightarrow$ Solution:

$(\omega t - \beta x)$: wave travelling in x dirn.

$-\beta x$: Shows +ve dirn. of field.

• For x dirn.:

$$\frac{E_x}{H_y} = -\frac{E_y}{H_x} = \eta$$

$$\therefore E_x = \eta H_y \quad \text{and} \quad E_y = -\eta H_x$$

$$\text{Now, } H_x = -\frac{E_y}{\eta} = \frac{-E_y}{120\pi}$$

$$= \frac{-10^{-3}}{120\pi} \sin(\omega t - \beta x)$$

$$\therefore H_x = -2.65 \sin(\omega t - \beta x)$$

* Relation betn. E and H for different dirns.

$\rightarrow$ If the wave is in +ve x -dirn. | y -dirn. | z -dirn.

$$1) E_x = H_x = 0$$

$$E_y = H_y = 0$$

$$E_z = H_z = 0$$

$$2) \eta = \frac{E_y}{H_z} = -\frac{E_z}{H_y}$$

$$\eta = \frac{E_x}{H_z} = -\frac{E_z}{H_x}$$

$$\eta = \frac{E_y}{H_x} = -\frac{E_x}{H_y}$$

$\rightarrow$ -ve x -dirn.

$$1) \eta = -\frac{E_y}{H_z} = \frac{E_z}{H_y}$$

-ve y -dirn.

$$\eta = -\frac{E_x}{H_z} = \frac{E_z}{H_x}$$

-ve z -dirn.

$$\eta = -\frac{E_x}{H_y} = \frac{E_y}{H_x}$$

- Let $\bar{a}_E$: unit vector in dirn. of $\bar{E}$
- $\bar{a}_H$: unit vector in dirn. of $\bar{H}$
- $\bar{a}_B$: ut. vector in dirn. of propagation of wave

then, $\bar{a}_B = \bar{a}_E \times \bar{a}_H$

- Let the wave in x -dirn. & $\bar{E}$ has E_y component
 $\therefore \bar{a}_B = \bar{a}_x$ also $\bar{a}_E = \bar{a}_y$

$\therefore \bar{a}_H = \bar{a}_z$ — (From x - y - z circle)

TYPE 1 - Free Space

- Q] A 150 MHz uniform plane wave in free space is travelling in $\bar{a}_x$ dirn., the $\bar{E}$ has max. amp. of $(200 \bar{a}_y + 400 \bar{a}_z)$ at $(P, -10, 30, -40)$ at $t=0$,
 $\rightarrow$ find phase const. η , velocity of phase (v), free space imp. (η), $E(x, y, z)$

$\rightarrow$ Solution:

- $\bar{a}_B = \bar{a}_x$

~~$\bar{a}_E = \bar{a}_z$~~

Now, $v = \frac{\omega}{\beta} = \frac{2\pi(150)(10^6)}{\beta} = 3(10^8) \text{ m/s}$

- $v = \frac{\omega}{\beta} = \frac{2\pi(150)(10^6)}{\beta} = 3(10^8)$

$\therefore \beta = \frac{\omega}{v} = \frac{2\pi(150)(10^6)}{3(10^8)} = 1\pi$

- Now, $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{1\pi} = 2 \text{ m}$

- $\eta = \sqrt{\mu\epsilon} = \sqrt{\frac{4\pi(10^{-7})}{8.85(10^{-12})}} = 120\pi$

- ∴ at $t=0$, amp. is maximum:
 ∴ the variation of E is in cosine manner and
 the wave is in x -dim.

$$\begin{aligned}\therefore \vec{E} &= (200\vec{a}_y + 400\vec{a}_x) \cos(\omega t + \beta x) \\ &= (200\vec{a}_y + 400\vec{a}_x) \cos(300\pi(10^6)t + -\pi x) \quad \text{V/m}\end{aligned}$$

TYPE 2

- Q] In a non-magnetic medium $\vec{E} = 50 \cos(10^9 t - 8x) \vec{a}_y + 40 \sin(10^9 t - 8x) \vec{a}_x$ (V/m), find ϵ_r and corresponding

→ Soln.

From given expression:

$$\omega = 10^9 \text{ rad/sec}, \quad \beta = 8$$

also, wave is propagating in x -dim.

- ∴ non magnetic medium:

$$\therefore \mu_r = 1$$

$$\text{we know, } \beta = \omega \sqrt{\mu \epsilon}$$

$$\therefore 8 = (10^9) \sqrt{(4\pi \times 10^{-7})(1)(8.85 \times 10^{-12}) \epsilon_r}$$

$$\therefore 64 = 10^{18} (4\pi \times 10^{-7})(8.85 \times 10^{-12}) \epsilon_r$$

$$\therefore \epsilon_r = 5.75$$

$$\text{Now, } \eta = \sqrt{\mu/\epsilon}$$

$$= \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}} = \frac{157}{\sqrt{5.75}} = 15.7$$

$$\therefore \eta = \frac{E_y}{H_x} \Rightarrow H_x = \frac{E_y}{\eta} \quad \text{Similarly, } H_y = \frac{-E_x}{\eta}$$

$$\begin{aligned}\text{Given, } \vec{E} &= 50 \cos(10^9 t - 8x) \vec{a}_y + 40 \sin(10^9 t - 8x) \vec{a}_x \\ &= \vec{E}_1 + \vec{E}_2\end{aligned}$$

$$\therefore \frac{E_1}{a_{\beta 1}} = \frac{50}{a_{\beta 1}} = \vec{a}_y$$

$$\frac{E_2}{a_{\beta 2}} = \frac{40}{a_{\beta 2}} = \vec{a}_x$$

$$\therefore H_2 = \frac{50 \cos(10^9 t - 8\pi)}{157} \bar{a}_y$$

$$= -0.31847 \cos(10^9 t - 8\pi)$$

Now, $\bar{H} = \bar{H}_1 + \bar{H}_2$

$$= H_1 \cos(10^9 t - 8\pi) \bar{a}_{H1} + H_2 \sin(10^9 t - 8\pi) \bar{a}_{H2}$$

$n = \frac{E_1}{H_1} = \frac{E_2}{H_2}$

$\therefore H_1 = 0.318, H_2 = 0.255 \text{ (A/m)}$

also, $\bar{a}_E \times \bar{a}_H = \bar{a}_B$

$\therefore \bar{a}_{B1} = \bar{a}_{E1} \times \bar{a}_{H1}$

$\bar{a}_{B2} = \bar{a}_{E2} \times \bar{a}_{H2}$

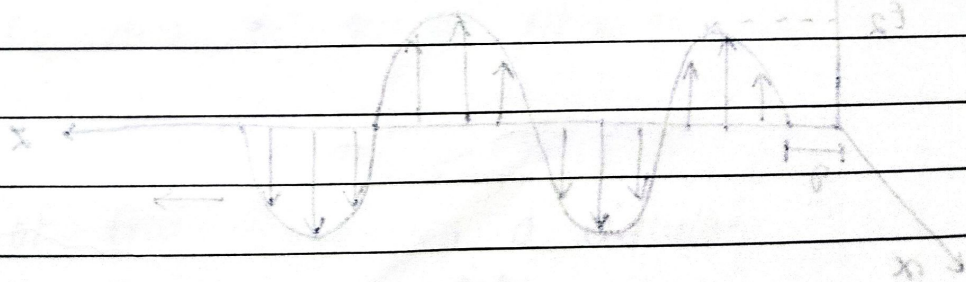
$\therefore \bar{a}_B = \bar{a}_y \times \bar{a}_{H1}$

$\bar{a}_B = \bar{a}_x \times \bar{a}_{H2}$

$\therefore \bar{a}_{H1} = \bar{a}_x$

$\therefore \bar{a}_{H2} = -\bar{a}_y$

$\therefore \bar{H} = 0.318 \cos(10^9 t - 8\pi) \bar{a}_x + 0.255 \sin(10^9 t - 8\pi) \bar{a}_y$



- * Polarization of wave: Time varying behaviour of $\vec{E}$ strength vector at some fixed pt. in space.

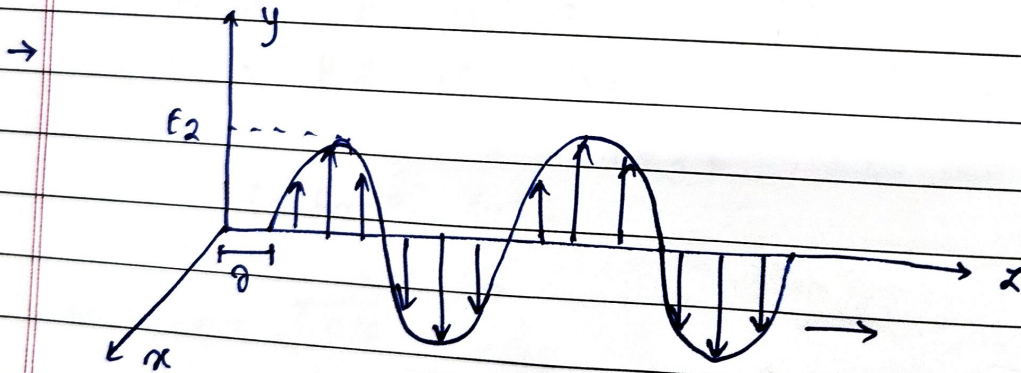
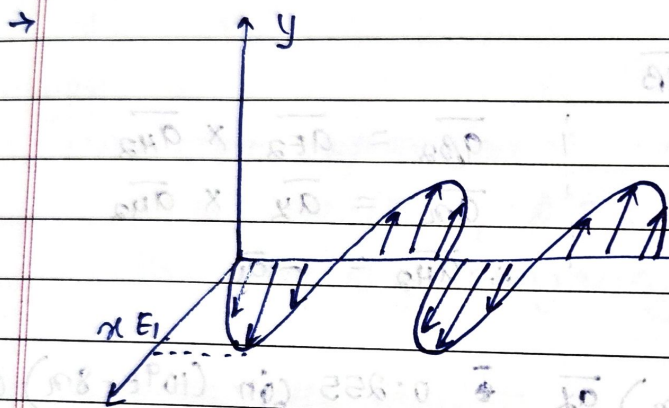
$$E_x = E_1 \cos(\omega t - \beta x)$$

$$E_y = E_2 \cos(\omega t - \beta x + \delta)$$

here, E_1 : amp. of wave in x -dim.

E_2 : " " " " y -dim.

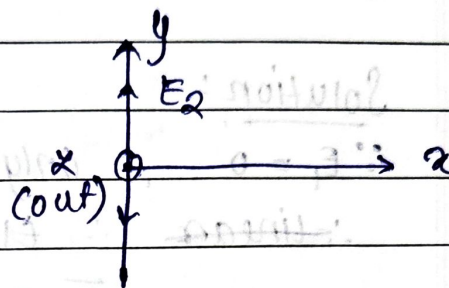
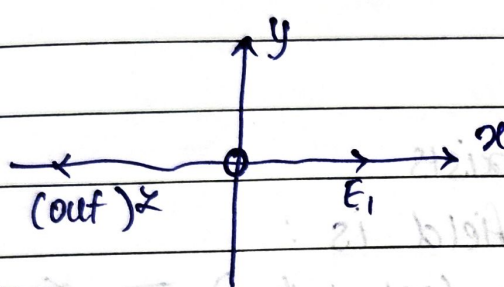
δ : phase $\angle$ betn. E_x & E_y .



- * Polarization:
- Linear
 - Circular (Mag. equal, $\angle$ different)
 - Elliptical

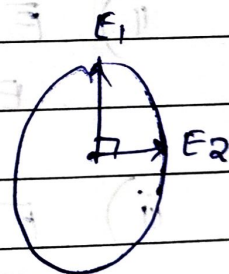
→ Linear polarization:

Time harmonic field is linearly polarized at given pt. in space, if $\vec{E}$ vector at that pt. always oriented along same straight line at any instant of time.



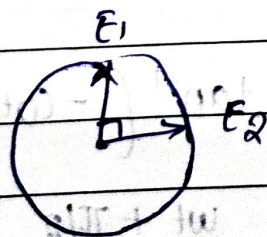
→ Elliptical:

- A wave is elliptically polarized, if the tip of the $\vec{E}$ vector traces an elliptical locus in space.
- Clockwise $\rightarrow$ RH polarized
Anti clockwise $\rightarrow$ LH " "
- E_1 & E_2 are 90° out of phase.



→ Circular:

- if tip of the $\vec{E}$ traces a circular locus.
- $E_1 = E_2$, phase diff. = 90° .



q) Consider the inst. $\vec{E}$ is given by
 $\vec{E} = E_1 \cos(\omega t - \beta x + \phi_x) \vec{a}_x + E_2 \cos(\omega t - \beta y + \phi_y) \vec{a}_y$

• Determine the poln. in $z=0$ plane if

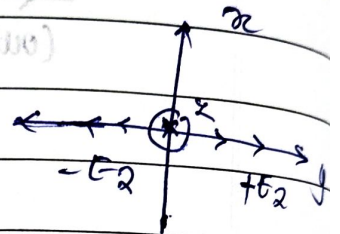
- i) $E_1 = 0$
- ii) $\phi_x = \phi_y = \phi$
- iii) $\phi_x = +\pi/2, \phi_y = 0$ & $E_1 = E_2 = A$
- iv) $\phi_x = 0, \phi_y = \pi/2$ & $E_1 = E_2 = B$
- v) $\phi_x = \pi/2, \phi_y = 0$ & $E_1 = A, E_2 = B$

→ Solution:

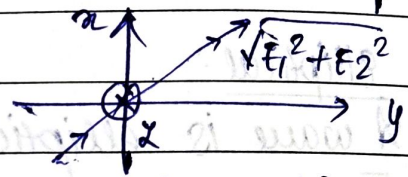
i) $\because E_1 = 0$ only E_2 exists

~~linear~~ $\therefore$ Electric field is:

$$\vec{E} = E_2 \cos(\omega t + \phi_y) \vec{a}_y$$



Straight line



$$\text{ii) } \vec{E} = E_1 \cos(\omega t + \phi) \vec{a}_x + E_2 \cos(\omega t + \phi) \vec{a}_y$$

$$\therefore E = \sqrt{E_1^2 + E_2^2} \cos(\omega t + \phi)$$

$$\text{iii) } \vec{E} = A \cos\left(\omega t + \frac{\pi}{2}\right) \vec{a}_x + A \cos(\omega t) \vec{a}_y$$

$$= -A \sin(\omega t) \vec{a}_x + A \cos(\omega t) \vec{a}_y$$

$$\therefore E = \sqrt{(-A \sin \omega t)^2 + (A \cos \omega t)^2} = A$$

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{-A \cos \omega t}{A \sin \omega t} \right) = \tan^{-1} (-\cot \omega t) \\ &= \omega t + \pi/2 \end{aligned}$$

$$iv) \quad \vec{E} = B \cos(\omega t) \vec{a}_m + B \cos\left(\omega t + \frac{\pi}{2}\right) \vec{a}_y$$

$$= B \cos(\omega t) \vec{a}_m - B \sin(\omega t) \vec{a}_y$$

$$\therefore E = \sqrt{\dots} = B$$

$$\therefore \phi = \tan^{-1} \int \omega t - \pi/2$$

v)

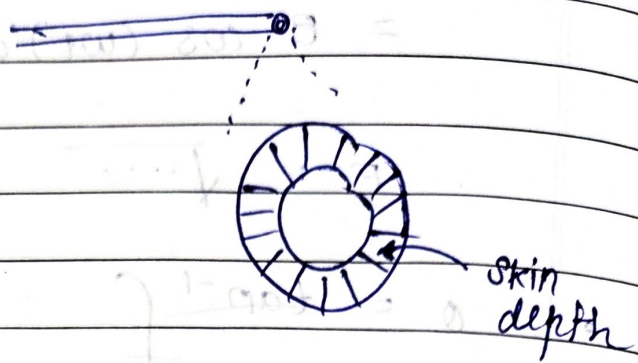
$$1 < 0.000005 = \frac{(2.11 \times 10^{-11}) (100 \times 10^2)}{\pi \times 8}$$

substituting value of ϵ_0 and μ_0 in the above equation

$$2.11 \times 10^{-11} \times 100 \times 10^2 = \pi \times 8$$

* Skin Depth / Depth of penetration:

- The depth in which the wave has been attenuated to $\frac{1}{e}$, i.e. approx. 37% of original value.



$$\delta = \frac{1}{\alpha} = \frac{1}{\left(\omega \sqrt{\frac{\mu \epsilon}{2}} \left(\sqrt{1 + \frac{\sigma^2}{\omega^2 \epsilon^2}} - 1 \right) \right)}$$

$$\therefore \delta = \sqrt{\frac{2}{\omega \mu \sigma}}$$

Q] $\sigma = 10^{-2} \text{ S/m}$, $\epsilon_r = 15$ & $\mu_r = 1$

Show that at freq. of 60 Hz, it can be characterized as good conductor.

calculate : i) Skin depth ii) Intrinsic α iii) propagation const.

→ Soln.:

$$\frac{\sigma}{\omega \epsilon} = \frac{(10^{-2})}{6\pi(60) \left(\frac{10^{-9}}{36\pi} \right) (15)} = 200000 \gg 1$$

$\therefore$ it is a good conductor.

i) $\delta = \sqrt{\frac{2}{\omega \mu \sigma}}$

$$ii) \cdot \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{j(2\pi)(60)(4\pi \times 10^{-7})}{10^{-2} + j(2\pi)(60)(\frac{10^{-9}}{36\pi})(15)}}$$

$$= 0.214 \angle 45^\circ$$

$$iii) \cdot \eta = \alpha + j\beta$$

For good conductor

$$\alpha = \beta = \frac{1}{\delta}$$

$$= 1.539 (10^{-3})$$

$$\therefore \eta = 1.539 (10^{-3}) + j(1.539 \times 10^{-3})$$

$$q) \sigma = 5(10^{-3}) \text{ S/m}$$

$\mu_r = 1$, $\epsilon_r = 8$. Can α be assumed zero at these frequencies.

→ Soln. :

For dielectric $(\sigma/\omega\epsilon) \ll 1$

assume $\frac{\sigma}{\omega\epsilon} \ll 1$

*

Poynting Theorem:

In a transmitter, the wave is generated by source travels to a distant receiving pt., whenever a wave travels from one pt. ^{another} ~~one~~, ~~xfer~~ of E. takes place.

The amp. of EM intensities ~~are~~ associated with travelling EM waves, this reln. is given by Poynting vector.

- Poynting Thm.
Vector prod. of $\vec{E}$ & $\vec{H}$ at any pt. is a measure of the rate

$$\vec{P} = \vec{E} \times \vec{H}$$

- The inst. P. flow contains stored E. & total P. flowing in the volume

$$\underbrace{\int_V (\vec{E} \cdot \vec{J}) dV}_{(i)} = - \frac{\partial}{\partial t} \underbrace{\int_V \left[\frac{\mu}{2} H^2 + \frac{\epsilon}{2} E^2 \right] dV}_{(ii)} - \underbrace{\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s}}_{(iii)}$$

i) $\int_V (\vec{E} \cdot \vec{J}) dV$: inst. P. dissipated

ii) Rate at which stored E. in volume is decreasing.
iii) + Rate at which E. is entering the volume from outside.